Provenance of the atomic correlation constant a_{corr} in the substrate framework: what is derived, what is conditional, and what remains open

Eliot Gable*
Independent Researcher
(Dated: November 1, 2026)

This is a provenance record for the correlation constant a_{corr} as the substrate corpus treats it, so that every statement can be checked against a file, a function and a commit. In the corpus $a_{\mathrm{corr}} = \epsilon_2 - \eta_2$ is the high-nuclear-charge, same-centre, two-electron second-order correlation coefficient of the Coulomb comparison model. $\eta_2 = \frac{9}{32}\ln\frac{3}{4} - \frac{13}{432}$ is derived in closed form; ϵ_2 is a published value, not a substrate result, so the number $a_{\mathrm{corr}} = -0.046663253999494962615$ is an imported value: a cross-check oracle, not a derivation. The corpus writes a_{corr} in two parts in several ways: a contact block fixed by the coincidence limit plus a non-contact remainder $\mathcal{R}_{\mathrm{tail}}$; an s-wave part (30.7 percent) plus a higher-partial-wave part (69.3 percent), of which the coincidence limit fixes only the large-l tail; and a charge-tail (single-source response) part plus a two-centre correlation part. The required route derives a_{corr} from the framework's two fixed inputs, the Born-Infeld Lagrangian and the closed-form particle expressions, and that derivation is open. On the present record the particle expressions enter only through the electron mass that sets the hartree and the Bohr radius, through the coupling that converts to electronvolts (conditional: the particle module marks its a_0 as not forward-closed), and through the Born-Infeld core scale relative to the atom; they do not supply the dimensionless value. The August element-generator scripts call their route forward, but it is not a derivation: it runs a variational Monte Carlo calculation of the ordinary two-electron Coulomb problem, an imported calculation of a differently defined quantity, installs the literature value 0.04667, and fits its occupation shape; its under-one-percent period-2 closure was superseded in September. None of the splits determines a_{corr}'s value from these premises. The test-charge kernel is not the equal-source interaction (exact contact ratio $4(\sqrt{2}-1)/3$). The two-centre part rests on the static two-source Born-Infeld energy. Its frozen superposition trial is closed in the sense of an exact linear differential equation with an exact algebraic-period germ, so it is computable by series continuation; the true static minimum is fenced but has no series; and the identification of either with the physical correlation of moving modes is a registered commitment that is not discharged. The physical a_{corr} remains open. We list the routes tried and the reasons each does not close, with their sources.

I. PURPOSE AND SCOPE

The owner's summary of this line of work is that a_{corr} "was derived into two parts with R_{tail} closed as a series". This paper sets out what the corpus actually establishes about that statement, with the evidence for each step, and reports where the statement aligns, partly aligns, or conflicts with the record. The owner's correction adds that a_{corr} "is not fixed merely from the lagrangian. It is also built based on the fixed closed form expressions for the particles"; Sec. IV traces how. It is a provenance document: it introduces no new derivation. Every numerical result is a macro generated by `papers/two_center_born_infeld/generate_numbers.py` from a corpus receipt at a recorded commit, or recomputed from a closed form; literals quoted from source files (such as the installed 0.04667) and exact constants written in closed form are given as they appear in their sources. That generator also verifies that each of the 60 formula strings transcribed into the two companion papers occurs verbatim in its cited source, and that each cited function exists. The references are origin/dev at `830cc5170df5`, the frozen-ODE branch at `3b8c2653996d`, the independent order-17 check at `cdd0f70ffa18`, the bond-research branch at `367287a67291` and the a_corr lane branch at `20cb3828b355` (full hashes in `papers/two_center_born_infeld/provenance_refs.json`). The static two-centre problem is treated in full in the companion paper [1]; here it appears only as one part of a_{corr}.

A. Standing context

The substrate framework is a Born-Infeld Lagrangian [2] describing an elastic solid with an excited $E_8 \times E_8$ dyad. $E_8 \times E_8$ is observationally overdetermined, D_{16} is excluded (`book/gr_qm_unification/sympy/substrate_theorems/`

* 10015996+egable@users.noreply.github.com

`rank16_observational_overdetermination.py`), and that includes its twist and spinor sector, its rank eight and its D_8 map. The framework has forward-derived quantization, entanglement and relativistic time dilation and is not classical. The particle mass equations are fixed closed forms in `book/scripts/substrate_mass_spectrum.py`. Particles are localized rotating wave modes of finite radius: two electrons in an atom are two lasting ripples of one medium, and their correlation is the discount in the energy of their joint, interfering pattern relative to the sum of the separate patterns. The Coulomb comparison model used below to define $a_{\rm corr}$ is an imported reference problem, not the substrate's account of that discount.

B. Status vocabulary

DEFINITION; DERIVED (forward from stated premises; the premises are named); CONDITIONAL (rests on a registered commitment); CERTIFIED-EXACT; DIAGNOSTIC (a computation inside an imported comparison model or with literature inputs); LITERATURE (an external value or theorem); IMPORTED (a value or calculation taken from outside the framework, such as a published value or a calculation of the ordinary two-electron Coulomb problem; usable only as a cross-check oracle, never as a derivation of $a_{\rm corr}$); OPEN; WITHDRAWN (asserted earlier, later retracted in the corpus). Table II gives each claim's source file, function where one exists, and the last commit touching it; a dagger marks a commit not contained in origin/dev.

II. WHAT $a_{\rm corr}$ IS IN THE CORPUS

A. Definition

For two electrons bound to one nucleus of charge Z in the nonrelativistic Coulomb problem, the ground energy has the large-Z expansion $E(Z) = -Z^2 + \frac{5}{8}Z + \epsilon_2 + O(1/Z)$ in hartree, and the Hartree-Fock energy has the same first two terms and second-order coefficient η_2. The corpus defines

$$a_{\rm corr} = \epsilon_2 - \eta_2 \tag{1}$$

(`atomic_correlation_constant_not_forced.py::verdict`; DEFINITION). The $1/Z$ expansion and the Hartree-Fock reference are perturbative constructs of the imported Coulomb model; in the substrate framework they serve only as a comparison problem, under the quarantine of perturbation theory as a method, and a substrate-native correlation must come from the moving-mode energy functional of Sec. V and the open items. It is a one-centre constant: it carries no dependence on a separation between centres, so it is not the two-centre correlation energy that a bond depth needs. The $O(Z)$ terms are equal because the exchange-lowered spatially antisymmetric partner of $1s^2$ is excluded in both curves (`pauli_shared_oz_term`), and the static monopole content $J = \frac{5}{8}Z$ cancels in the difference.

B. The closed part: η_2

The first-order orbital equation $(-\frac{1}{2}\nabla^2 - 1/\rho + \frac{1}{2})\phi_1 = -(V_C - \frac{5}{8})\phi_0$ with $V_C(r) = (1-(1+r)e^{-2r})/r$ is solved by $\phi_1 = f\phi_0$ in the Dalgarno-Lewis reduction [3], which turns it into a first-order equation for $w = f'$ whose solution regular at the origin is elementary. The energy integral then gives

$$\eta_2 = \frac{9}{32}\ln\frac{3}{4} - \frac{13}{432} = -0.111003175469655978435, \tag{2}$$

the logarithm arising by Frullani integration of the $1/r$ part of f' (`eta2_symbolic_derivation`; establish-or-bail checks of the defining equation, regularity, decay and the self-adjoint identity). It agrees with the published closed form of Loos and Gill [4], which is an oracle here. Status: DERIVED, inside the imported Coulomb operator, whose kinetic and potential terms are not derived from the Born-Infeld medium in this calculation.

C. The open part: ϵ_2

The corpus value $\epsilon_2 = -0.15766642946915094105$ is read from the publication of Wang [5] (`second_order_partial_waves.py::published_eps2`; LITERATURE). Hence the corpus number

$$\begin{aligned} a_{\rm corr} &= -0.15766642946915094105 - (-0.111003175469655978435) \\ &= -0.046663253999494962615, \qquad |a_{\rm corr}| = 0.0466632539995, \end{aligned} \tag{3}$$

is IMPORTED: it inherits the provenance of the published ϵ_2, serves as a cross-check oracle for the required derivation from the Born-Infeld Lagrangian and the particle equations, and is not a substrate prediction. The corpus reduces ϵ_2 to a sum over total-$L = 0$ partial-wave channels. The s-wave particular solution $\frac{5}{8}r_> - \frac{3}{8}\ln r_> + \frac{3}{16 r_>}$ is exact; the large-l tail follows the Schwartz law $-\frac{45}{256}(l+\frac{1}{2})^{-4} + \frac{225}{1024}(l+\frac{1}{2})^{-6} + \dots$ [6], whose sum is elementary through half-integer Hurwitz values; the matching data at the crossover shell $r_1 = r_2$ are fixed in closed form by a field junction. The crossover correction, however, is not in the five-dimensional elementary-plus-exponential-integral homogeneous space of the channel operator, so the s-wave channel function is not of that elementary form and $\epsilon_2(0)$ does not close by this assembly; numerically $\epsilon_2(0) = -0.125334(1)$ (`eps2_swave_crossover_is_non_elementary`). This is a non-membership result for one assembly, not an impossibility proof for a closed form. Status of a closed form for ϵ_2: OPEN.

III. THE TWO-PART DECOMPOSITIONS

The corpus writes $a_{\rm corr}$ as two parts in three different senses. They must not be conflated, and the symbol $R_{\rm tail}$ carries different meanings across them (written $\mathcal{R}_{\rm tail}$ here for the remainder of the first split).

A. Contact block plus non-contact remainder

Write the target as a global integral $a_{\rm corr} = \int_0^\infty G(r_{12})\, w(r_{12})\, dr_{12}$ of a two-centre field-energy density against a pair measure, choose an overlap radius r_* and a finite set of retained coincidence exponents p_k with coefficients G_k of the small-r expansion of G, and split

$$a_{\rm corr} = \sum_k \mu_k G_k + \mathcal{R}_{\rm tail}, \qquad \mu_k = \int_0^{r_*} r^{p_k} w\, dr, \qquad \mathcal{R}_{\rm tail} = \mathcal{R}_{\rm loc} + \mathcal{R}_{\rm out}, \tag{4}$$

$$\mathcal{R}_{\rm loc} = \int_0^{r_*} \Big(G - \sum_k G_k r^{p_k}\Big) w\, dr, \qquad \mathcal{R}_{\rm out} = \int_{r_*}^\infty G\, w\, dr. \tag{5}$$

This is an exact identity for any integrable G and w and any finite retained set. The local remainder $\mathcal{R}_{\rm loc}$ is the part of the overlap integral that the retained contact coefficients do not carry, and the corpus does not establish its value. Equality of G with its retained expansion on $[0, r_*]$ is sufficient for $\mathcal{R}_{\rm loc} = 0$ but not necessary: for $G = r^3 + r^{10}(r - 11/12)$, which is positive on $(0, 1]$, with $w = 1$, $r_* = 1$ and retained exponents $3, \dots, 9$, the retained expansion is r^3 and $\mathcal{R}_{\rm loc} = 0$, while the residual at $r = 1/2$ is $-5/12288$ (checked exactly by the generator). It is not negligible in general: for $G = r^{10}e^{-r}$, $w = 1$, $r_* = 1$ and retained exponents $3, \dots, 9$, every retained G_k is zero while $\mathcal{R}_{\rm loc} = 10! - 9864101/e = 0.0364613 > 0$ (checked symbolically by the generator). The source script writes the remainder as the outer integral alone (`acorr_proximity_cap_rank_analysis.py`); its rank calculation treats $\mathcal{R}_{\rm tail}$ as one open unknown, so it holds unchanged for the remainder defined here. Matching the global density to the local coincidence hole, $G_k = L_k$, is a square system: one equation and one open contact coefficient per retained order, with no surplus equation, so the match fixes the retained contact coefficients and places no constraint on $a_{\rm corr}$ (`substrate_correlation_coincidence_match_underdetermines`; DERIVED). A Born-Infeld proximity cap at a minimum separation $r_{\rm min}$, written as a condition on the same retained coefficients, behaves the same way in that finite contact-coefficient model: symbolically and over 200 random instances its row lies in the span of the match rows, its projection on the $(a_{\rm corr}, \mathcal{R}_{\rm tail})$ columns vanishes, and the $(a_{\rm corr}, \mathcal{R}_{\rm tail})$ subsystem stays rank one in two unknowns (`substrate_correlation_proximity_cap_is_contact_local_no_rank`; DERIVED for that finite model). Substituting the match into the cap row leaves the consistency condition $\sum_k L_k r_{\rm min}^{p_k} = 0$ on $r_{\rm min}$. The corpus does not show that this condition has a positive root, or that a root is unique; its function leaves the corresponding result unset (`cap_consistency_fixes_r_12_min_not_a_corr`: `None`), so the condition is recorded as a consistency condition,

not as a determination of $r_{\min}$. The split is therefore a bookkeeping identity. What it shows is narrower than a general locality theorem: conditions on the retained contact coefficients alone (the match and the hard or soft cap) do not constrain $a_{\rm corr}$. It does not exclude a local boundary condition that, together with a governing global field equation for G, fixes $a_{\rm corr}$; the corpus function states that limit itself and keeps that outcome open. The remainder $\mathcal{R}_{\rm tail}$ contains both the local Taylor remainder and the outer integral, and both are open.

B. s-wave plus higher partial waves

With literature per-l increments of the Coulomb problem (Carroll, Silverstone and Metzger [7]) the corpus finds the s-wave correlation $\epsilon_2(0) - \eta_2 = -0.014326825$, 30.7 percent of $a_{\rm corr}$, and 69.3 percent in $l \geq 1$ (`acorr_partialwave_decomposition.py`; DIAGNOSTIC). The coincidence limit $r_{12} \to 0$ is the large-l limit: it reproduces the shared cusp law, whose sum from $l = 0$ is $+11.23$, nowhere near ϵ_2, while most of $a_{\rm corr}$ sits in the low-l head that the coincidence data never reach. The resisting content is the short-range Kato coalescence cusp [8], whose branch point gives the partial-wave expansion its power-law tail.

C. Charge-tail part plus two-centre correlation

a. Charge tail. The single-source Born-Infeld potential and the linear response of an infinitesimal test charge are evaluated in `acorr_charge_tail_dependency_FINDINGS.md`. For $s \geq 0$ and core scale $r_0 > 0$, the substitutions $r = r_0(1-t)/t$, $t = au$, $a = r_0/(s + r_0)$ give

$$U(s) = \frac{1}{s + r_0} \int_0^1 \frac{du}{\sqrt{(1-au)^4 + (au)^4}}, \tag{6}$$

and, for $s > 0$ (where the bounds are defined),

$$\frac{1}{s} - \frac{r_0^4}{10 s^5} \leq U(s) \leq \frac{1}{s}, \tag{7}$$

with the quartic in $[1/8, 1]$ on the fixed interval, which removes a scale-dependent quadrature failure (DERIVED, scalar route only; landed numerical-domain component). The same record states the scope plainly: the mechanical displacement, strain, scalar potential and conjugate flux are distinct types, a scalar $1/r$ potential does not identify a compatible mechanical strain, and no physical $a_{\rm corr}$ follows. The decisive check is at contact. By displacement-preserving dilation a source of charge q has energy $Kq^{3/2}$ with $K = \Gamma(1/4)^2/(6\sqrt{\pi}) = 1.236049784867581$, so the interaction of two finite sources at contact is $V(q_1, q_2; 0) = K[(q_1+q_2)^{3/2} - q_1^{3/2} - q_2^{3/2}]$, while the test-charge coefficient is its one-sided derivative. For two unit sources

$$\frac{V(1,1;0)}{U(0)} = \frac{4(\sqrt{2}-1)}{3} = 0.5522847498307934 \neq 1, \qquad U(0) - V(1,1;0) = \Big(\frac{7}{2} - 2\sqrt{2}\Big) K = 0.8300975079803747, \tag{8}$$

(`bi_test_charge_pair_energy.py::contact_interaction_certificate`; CERTIFIED-EXACT). The isolated-source test-charge potential is therefore not the equal-source interaction already in the static point-source diagnostic, and the partial-wave calculations that used it as a pair kernel are comparison calculations whose transfer to two finite sources is not established. The static linear far-field strain of a spherical source is pure longitudinal and pure monopole; inside the Coulomb comparison operator its content is the $l = 0$ direct term that cancels in $a_{\rm corr}$, and the retarded magnetic wake enters at $O(\alpha^2) \sim 5 \times 10^{-5}$ (`substrate_correlation_strain_tail_and_wake_not_the_resisting_piece`; DIAGNOSTIC). This does not exclude wake, twist or retardation contributions to the physical energy of finite moving modes; the framework holds that those effects are always present, and their size must be computed, not assumed.

b. Two-centre correlation. The second part is the energy of the joint two-source field, which is where Born-Infeld nonsuperposition acts. The corpus registers its identification with $a_{\rm corr}$ as the commitment `rtail_two_center_m11_physical_identification`: mapping the static Euclidean two-centre energy to the correlation of moving atomic modes requires the finite moving-mode reduction (observer clock and measure, retarded wake, angular momentum and twist, interference), which is not built (CONDITIONAL). The M11 route validation, which tried to compute $a_{\rm corr}$ from the true two-electron field minimizer, returns open with that blocking antecedent named, `full_elastic_observer_energy is None`, together with the missing physical two-electron bound mode; it also records that softening the interaction alone, with the nonrelativistic framework fixed, shifts $a_{\rm corr}$ by $O(r_0^2) \sim 10^{-9}$ at atomic scale, so the leading part reduces to the Coulomb $\epsilon_2 - \eta_2$ unless the moving-mode terms change it (`acorr_m11_true_minimizer_route_validation.py`; OPEN).

IV. HOW THE CLOSED-FORM PARTICLE EXPRESSIONS ENTER, AND THE ELEMENT-GENERATOR ROUTE

The owner's correction is that $a_{\rm corr}$ "is also built based on the fixed closed form expressions for the particles". The owner has also ruled that $a_{\rm corr}$ must come from the Born-Infeld Lagrangian and the particle equations and must not be imported: a Monte Carlo calculation of the ordinary two-electron Coulomb problem is not that route. That derivation is the required route, and it is OPEN. The corpus confirms that the particle equations of `book/scripts/substrate_mass_spectrum.py` enter the $a_{\rm corr}$ calculations through units and scale, and the element-generator scripts of August 2026 present a route that they call forward but that is not a derivation. This section traces each step to its source, with its status. All steps below sit inside the same nonrelativistic Coulomb comparison problem as Sec. II unless stated otherwise.

a. Step P1: the electron mass. Under the module's default anchor (`default='structural'`), the electron rest energy is the fixed closed form on the ϕ/E_8 ladder times the structural scale $h_{\rm eff}$ (`substrate_mass_spectrum.py::derive_constants`). The electron mass sets the hartree and the Bohr radius, the units in which $a_{\rm corr}$ is a pure number. Status: FIXED CLOSED FORM, a framework premise; it fixes the unit, not the dimensionless value $a_{\rm corr}$.

b. Step P2: the coupling and the energy unit. The element generator converts hartree to electronvolts with $E_h = \alpha_{\rm sub}^2 m_e c^2$, $\alpha_{\rm sub} = \lambda_{C,e}/(2\pi a_0)$, taken from the same module (`forward_p_row_generator.py::hartree_eV`); if that import fails it silently falls back to the measured hartree. The module's own status block marks a_0 and the Rydberg energy as not forward-closed: they inherit an α that consumes imported QFT corrections, with the Born-Infeld $\alpha(0)$ closure pending. Status: CONDITIONAL. This step sets the eV value of $a_{\rm corr}$, not its value in hartree.

c. Step P3: the Born-Infeld core scale relative to the atom. The size of the Born-Infeld softening of the pair interaction relative to the Coulomb value is set by the core radius r_0 in Bohr units, which uses the identification of the single-source energy with the electron rest energy. The M11 validation records that softening alone shifts $a_{\rm corr}$ by $O(r_0^2) \sim 10^{-9}$ at atomic scale (Sec. III C). Status: CONDITIONAL on that identification and on the fixed nonrelativistic framework.

d. Step P4: the r_{12} Jastrow route. `derive_pair_correlation_constant_forward.py` (with `derive_helium_correlation_forward.py`) evaluates the point-Coulomb Hamiltonian, `pot = -Z / r1 - Z / r2 + 1.0 / r12` with the Schrödinger kinetic operator, for the trial $e^{-\zeta(r_1+r_2)}(1 + cr_{12})$ by fixed-seed variational Monte Carlo, defines $E_{\rm corr}(Z) = -(Z-5/16)^2 - \min E(\zeta, c; Z)$, and extrapolates in Z. Its recorded values are $Z = 2$: 0.0435, $Z = 4$: 0.0467, $Z = 8$: 0.0472, $Z = 16$: 0.0497 hartree; the script's own least-squares extrapolation $a + b/Z$ over $Z = 4, 8, 16$ gives 0.04995, consistent with its commit message ("extrapolating to ∼0.05 Ha"), while `FORWARD_ELEMENT_GENERATOR.md` reports $a \approx 0.047$. No particle expression and no Born-Infeld term enters this calculation; its "tension cusp" is the Kato cusp of the Coulomb operator. Its reference is the screened single-exponent energy, whose Z^0 term is $-25/256 = -0.09765625$, not the Hartree-Fock η_2; relative to that reference the exact Z^0 correlation is $|\epsilon_2 + 25/256| = 0.0600102$, so the one-parameter Jastrow estimate is a variational partial recovery of a differently defined quantity, and its closeness to $|a_{\rm corr}| = 0.0466632539995$ does not identify the two. The check passes when the result lies within 0.008 of the fitted generator value 0.04738; Monte Carlo uncertainty is not quantified. Status: IMPORTED, a cross-check calculation of the ordinary Coulomb problem only; it is not a forward derivation, not substrate-native and not the defined quantity, and by the owner's ruling it cannot supply $a_{\rm corr}$.

e. Step P5: the value actually installed. The element generators install the literal 0.04667 (`forward_missing_terms_generator.py`; `LEGACY_DIAGNOSTIC_A_CORR=sp.Rational(4667,100000)` in `forward_generator_complete.py`). That number is the He-like high-density limit, the imported corpus value $|\epsilon_2 - \eta_2|$ of Sec. II, whose ϵ_2 is a published value. Status: IMPORTED (a literature value installed); the current files label it a legacy diagnostic, and it serves only as a cross-check oracle.

f. Step P6: the occupation shape and the 3:4 ratio. The missing-terms polynomial $a\,k(3k-11)/8$ with $b/a = -3/4$ was obtained by a least-squares fit of the six period-2 residuals (`forward_missing_terms_generator.py`: "This script DISCOVERS the RATIO pattern (fits to expose the 3:4 lock and the single constant a)"). The identification of 3/4 with the spin-$\frac{1}{2}$ Casimir used in the particle sector is recorded as a lead "TO CONFIRM", and the particle-generator origin of the value 0.0467 as open. The September record rejects 1/21 and 3/64 as exact values of $|a_{\rm corr}|$: they differ from it by +2.05 and +0.45 percent, outside its error bar. Status: shape and ratio FITTED; particle-sector origin OPEN.

g. Step P7: the September corrections. The claim that this generator plus one constant closes period 2 to under one percent with zero fits was superseded in September 2026. The legacy optimizer minimized a removed-orbital expression, not the atom's common occupied energy; the exact derivative defects are $[0, -21/64, -63/64, -585/256, -921/256, -1341/256]$ for $k = 1, \ldots, 6$ (`atomic_generator_energy_consistency_FINDINGS.md`), and selecting the common energy fixes that objective (`atomic_p_relaxed_energy_FINDINGS.md`); the two correction commits are in Table III. The candidate shape is also not what a single-amplitude pair correction produces (`atomic_pshell_pair_structure_FINDINGS.md`). A probe by the ionization-generator lane (`40cd2c13f73b`, unreviewed) reproduces the

September table and gives B to Ne at -16.1 to -53.6 percent with the corrected objective, and finds that the fitted shape does not transfer to the corrected generator. Status: the under-one-percent closure is WITHDRAWN.

h. Relation to the decomposition of Sec. II and to M11. The installed number and the corpus definition are the same quantity, $a_{\rm corr} = \lim_{Z\to\infty}(E_{\rm exact} - E_{\rm HF}) = \epsilon_2 - \eta_2$, whose value is known numerically from published work. The Jastrow Monte Carlo calculation addresses a related but differently defined quantity of the same Coulomb model, consistent in size and not in conflict with the September identification, which kept the stored value and found no closed form. Both are imported cross-checks inside the Coulomb comparison problem, and neither is a derivation; both lack the M11 step, the moving-mode two-electron energy functional of the Born-Infeld medium. The required route builds $a_{\rm corr}$ from the Born-Infeld Lagrangian and the particle equations; the particle equations fix the units and the atomic scale through which that functional would be expressed, and on the present record they do not supply the dimensionless value of $a_{\rm corr}$. That route is OPEN.

V. $R_{\rm tail}$ AND ITS SERIES

In the files named `rtail_*` the symbol $R_{\rm tail}(d)$ denotes the static two-source Born-Infeld energy at separation d, either of the true minimizer $E_{\rm min}(d)$ or, in older files, of the frozen superposition trial $W_{\rm sup}(d)$ (the reduction once labelled the true $R_{\rm tail}$ was relabelled as the frozen trial; see the timeline). The companion paper [1] gives the full account; the results that bear on the owner's statement are these.

1. The frozen trial $W_{\rm sup}$ satisfies an exact linear differential equation of order 18 with integer polynomial coefficients, certified by exact divergence identities (replayed for this record), and a smaller order-17 annihilator has independent source clearance on its branch. The physical branch has order at least 4 and is not algebraic. An exact germ of eighteen algebraic periods at $d = 1$ selects the solution, so $W_{\rm sup}$ is computable to arbitrary precision by series continuation once the germ is enclosed to that precision; a recorded continuation step agrees with independent quadrature to 1.07×10^{-81}, and an unreviewed multi-step continuation to 2.1×10^{-53} or better on $1.25 \le d \le 4$ (observed agreements, not error bars). In this precise sense the frozen trial is "closed as a series".

2. The frozen trial is a strict upper bound on the static energy, never the energy itself. For the true minimum the corpus has certified fences (the contact force 1/2 from a contact enclosure, the interval $0.63726946 \le V(1) \le 0.64626656$, a dual lower bound) but no differential equation and no series; its large-separation law is conditional on a registered commitment.

3. Neither static object has been identified with the physical two-mode correlation (the commitment above), and neither is the remainder $\mathcal{R}_{\rm tail}$ of the first split, which is an integral of a density that the corpus has not constructed.

The former claim that the physical two-centre energy is a proved non-elementary order-four Picard-Fuchs period is WITHDRAWN: `rtail_known_problems_context.md` states that the former claim is withdrawn and that the checked algebra concerns the frozen trial. The derivations README at the recorded origin/dev commit still carried the older "now PROVEN NON-ELEMENTARY" paragraph; the corpus commit that accompanies this paper replaces it with the current status, and corrects the two-centre synthesis module, which had called the true large-separation law exact.

VI. ROUTES TRIED AND THEIR OUTCOMES

TABLE I: Routes to $a_{\rm corr}$ recorded in the corpus.

route	outcome	source
Constant installed in the forward element generator (August)	imported: the original 2p diagnostic installs 0.04667 hartree as an input; it has no forward derivation, and a later commit separated it from the cross-shell correlation E_c	a_corr lane journal; timeline
ϵ_2 by Green's-function assembly	crossover correction outside the elementary plus Ei span; $\epsilon_2(0)$ does not close by this assembly (non-membership)	`eps2_swave_crossover_is_non_elementary`
Rigid limit versus deformed metric	the lapse rationalizes the source but the acoustic-measure root cancels it; the universal cancellation was later withdrawn and the observer measure must be derived	`substrate_correlation_hole_is_a_deformed_metric_problem`

route	outcome	source
Global field-energy route	the saturation root resums globally and the single-centre building blocks are $\Gamma(1/4)$-clean; the correlation functional and its field-energy map remain open	`substrate_correlation_global_field_energy_route`
Coincidence match	square system; under-determines a_{corr}	`substrate_correlation_coincidence_match_underdetermines`
Static strain tail and wake	inside the comparison operator: the static tail is $l = 0$ and cancels, the wake is $O(\alpha^2)$; not a statement about finite moving modes	`substrate_correlation_strain_tail_and_wake_not_the_resisting_piece`
Cuspless test-charge kernel	comparison calculation; the kernel is not the equal-source interaction	`substrate_native_cuspless_correlation_constant`
Proximity cap	adds no rank in the finite contact-coefficient model; its consistency condition on $r_{\min}$ has no proved root	`substrate_correlation_proximity_cap_is_contact_local_no_rank`
Order-four Picard-Fuchs closure of R_{tail}	withdrawn; downgraded to a rank-six upper bound with open obligations, then superseded by the exact order-18 equation for the frozen trial	`rtail_known_problems_context.md`
M11 true-minimizer route	open; the moving-mode two-electron functional and the physical bound mode are missing	`acorr_m11_true_minimizer_route_validation.py`
Element-generator r_{12} Jastrow route (August)	imported: Monte Carlo of the ordinary Coulomb problem for a screened-reference quantity, extrapolated near 0.05, a cross-check and not a derivation; generator installs the literature 0.04667 and a fitted shape; under-one-percent closure withdrawn in September	Sec. IV
Harmonium pair law	not a forced universal constant	`harmonium_correlation_not_forced.py`

The module title `atomic_correlation_constant_not_forced.py` records the cumulative result: its verdict exports `"a_corr": None` and `"physical_generator_complete": False`. "Not forced" there means that no route recorded so far forces a value; it is not a proof that no route can.

VII. WHAT IS CERTIFIED, AND TO WHAT PRECISION

- Exact: η_2 in closed form; the contact ratio $4(\sqrt{2}-1)/3$; the endpoint energies $2K$ and $2^{3/2}K$ of the static two-source problem; the order-18 annihilator of the frozen trial and its indicial data; the frozen large-separation law $W_{\mathrm{sup}} = 2K + 1/d - (3K/4)/d^4 + o(d^{-4})$; the lower bound 4 on the frozen trial's order and its nonalgebraicity.
- Interval-certified: the true static minimum's contact enclosure on $0 \le d \le 1/2$ and its interval at $d = 1$.
- Observed (agreement of routes, not error bars): the frozen trial's values and continuation agreements; the numerical $\epsilon_2(0)$.
- Imported (cross-check oracles only, not derivations): ϵ_2 and hence the value of a_{corr}, the per-l increments used in the partial-wave split, the generator literal 0.04667, and the Coulomb-problem Monte Carlo estimate.
- Fitted: the period-2 occupation shape and its 3:4 ratio. Withdrawn: the period-2 under-one-percent closure.
- Fixed closed forms entering as units and scale: the electron mass (premise); the coupling for the eV conversion (conditional).

VIII. ASSESSMENT OF THE OWNER'S STATEMENT

The statement partly aligns with the record. It aligns in that the corpus does split a_{corr} into two parts, in each of the three senses above, and in that the frozen two-centre trial energy is closed in the sense of an exact differential equation with an exact germ, computable to arbitrary precision by series continuation. It conflicts in three places.

First, a_{corr} must be built from both the Lagrangian and the closed-form particle expressions, as the owner states, but that derivation is open, and neither input, nor any split, determines its dimensionless value on the present record (Sec. IV): the contact-plus-remainder split is a bookkeeping identity whose remainder, local and outer, is unconstructed, the partial-wave split uses published Coulomb values, and the charge-tail-plus-two-centre split rests on an identification that is a registered, undischarged commitment. Second, what is closed as a series is the frozen superposition trial, an upper bound, not the true static minimum and not the correlation of moving modes. Third, the corpus value of a_{corr} inherits the published ϵ_2, and the Monte Carlo estimate is a calculation of the ordinary Coulomb problem; both are imported cross-check oracles, not substrate results and not derivations. The physical a_{corr} is open.

IX. WHAT REMAINS OPEN

1. The required route: a_{corr} derived from the Born-Infeld Lagrangian and the closed-form particle equations, with the imported $\epsilon_2 - \eta_2$ value and the Coulomb-problem Monte Carlo estimate used only as cross-check oracles.
2. The physical moving-mode two-electron correlation functional, with its observer clock and measure (by the owner's ruling, one global tick and the actual local voxel volume), retarded strain field, motion wake, twist and interference, and its identification with a_{corr}; and the physical bound mode it requires.
3. The charge sector of the master Lagrangian's cross term (phase-carrying mode class, source coupling, $1/r$ tail and Maxwell regime), which the a_corr lane orders before the bound mode.
4. Whether the particle sector fixes the dimensionless value of a_{corr} (the 3/4 Casimir lead and the origin of the number), and a forward-closed coupling for the eV conversion.
5. A closed form, or a proof of impossibility, for ϵ_2 and for $\epsilon_2(0)$.
6. The remainder $\mathcal{R}_{\mathrm{tail}}$, local Taylor remainder and outer integral, as actual integrals of a constructed density; and whether the cap consistency condition $\sum_k L_k r_{\mathrm{min}}^{p_k} = 0$ has a positive root, and a unique one.
7. For the static two-centre energy: the items listed in [1], including the minimal order of the frozen trial and any series or equation for the true minimum.
8. Verification of corpus citation fields not checked for this record, for example the forcing node's statement of what Carley and Kiessling establish about the linear Bopp-Podolsky potential.

Appendix A: Claims, status and sources

Paths are relative to `book/gr_qm_unification/sympy/`.

TABLE II: Claims of this paper with status and source.

id	claim	status	source	commit
A1	a_corr is defined as eps_2 - eta_2 of the Coulomb comparison model (high-Z, same centre, two electrons)	DEFINITION	`substrate_theorems/atomic_correlation_constant_not_forced.py::verdict`	`56d7c16cfe`
A2	eta_2 = (9/32)ln(3/4) - 13/432 by the Dalgarno-Lewis reduction	DERIVED (imported Coulomb operator)	`substrate_theorems/atomic_correlation_constant_not_forced.py::eta2_symbolic_derivation`	`56d7c16cfe`
A3	eps_2 = published value (Wang 2013), not a substrate result; with it the a_corr value is a cross-check oracle	IMPORTED (literature; cross-check oracle only)	`imported_theorems/second_order_partial_waves.py::published_eps2`	`56d7c16cfe`
A4	s-wave and l>=1 split of a_corr from literature per-l increments	DIAGNOSTIC (literature inputs)	`derivations/acorr_partialwave_decomposition.py`	`674a28b5fd`
A5	the s-wave crossover correction lies outside the elementary plus Ei span (non-membership only)	DERIVED (conditional)	`substrate_theorems/atomic_correlation_constant_not_forced.py::eps2_swave_crossover_is_non_elementary`	`56d7c16cfe`

id	claim	status	source	commit
A6	coincidence match is a square system on the retained contact coefficients; it fixes them, not a_corr	DERIVED	substrate_theorems/atomic_correlation_constant_not_forced.py::substrate_correlation_coincidence_match_underdetermines	56d7c16cfe
A7	proximity cap rows add zero rank; (a_corr, remainder) block stays rank 1 in 2 unknowns	DERIVED (finite contact-coefficient model only)	derivations/acorr_proximity_cap_rank_analysis.py	56d7c16cfe
A8	test-charge contact ratio V(1,1;0)/U(0) = 4(sqrt2-1)/3, not 1	CERTIFIED-EXACT	substrate_theorems/bi_test_charge_pair_energy.py::contact_interaction_certificate	b436d403d6
A9	scalar point-source potential: exact substitution, quartic range [1/8,1], Coulomb bracket	DERIVED (scalar route only)	derivations/acorr_charge_tail_dependency_FINDINGS.md	cfa43b6430
A10	static linear strain tail is l=0 and the wake is O(alphaˆ2) inside the Coulomb comparison only	CONDITIONAL DIAGNOSTIC	substrate_theorems/atomic_correlation_constant_not_forced.py::substrate_correlation_strain_tail_and_wake_not_the_resisting_piece	56d7c16cfe
A11	static two-centre energy to a_corr identification	COMMITMENT (M11-gated)	substrate_theorems/rtail_two_center_synthesis.py::two_center_synthesis	56d7c16cfe
A12	M11 true-minimizer route: physical moving-mode functional and bound mode missing	OPEN	derivations/acorr_m11_true_minimizer_route_validation.py	56d7c16cfe
A13	former order-4 Picard-Fuchs non-elementarity claim	WITHDRAWN	derivations/rtail_known_problems_context.md	745293d9c5
A14	a_corr derived from the Born-Infeld Lagrangian and the particle equations (the required route)	OPEN	substrate_theorems/atomic_correlation_constant_not_forced.py::verdict	56d7c16cfe
A15	electron mass from the fixed closed-form particle equations (default structural anchor)	FIXED CLOSED FORM (framework premise)	book/scripts/substrate_mass_spectrum.py::derive_constants	d90bb8280a
A16	energy unit alphaˆ2 m_e cˆ2 that expresses a_corr in eV, from the particle module	CONDITIONAL (module marks a_0 and Ry not forward-closed)	book/scripts/forward_p_row_generator.py::hartree_eV	de463350ee
A17	r12 Jastrow VMC 'forward a': Monte Carlo of the ordinary Coulomb problem, screened reference, extrapolated	IMPORTED (cross-check only; not a derivation)	book/scripts/derive_pair_correlation_constant_forward.py::corr_at_Z	79ab032f5a
A18	b/a = -3/4 identified with the spin-1/2 Casimir of the particle sector	OPEN (lead, to confirm)	book/scripts/FORWARD_ELEMENT_GENERATOR.md	7fd6e5c283
A19	element generator installs a_corr = 4667/100000 as a legacy diagnostic literal	IMPORTED (literature value installed)	book/scripts/forward_generator_complete.py::legacy_orbital_ie_diagnostic	82ead9b8d9
A20	occupation shape k(3k-11)/8 and b/a = -3/4 found by least-squares fit of the period-2 residual	FITTED	book/scripts/forward_missing_terms_generator.py	c0610401eb
A21	legacy optimizer minimized a removed-orbital expression, not the atom's energy (September correction)	CERTIFIED-EXACT (defect)	derivations/atomic_generator_energy_consistency_FINDINGS.md	de463350ee
A22	period 2 under 1% with zero fits superseded; corrected objective gives B to Ne -16.1 to -53.6 percent	UNREVIEWED (other lane probe)	work_journal/ionization-generator-period3-open.md	40cd2c13f7†
A23	proximity cap consistency sum_k L_k r_minˆp_k = 0 is a condition on r_min; a positive root and its uniqueness are not proved	OPEN (result left unset)	substrate_theorems/atomic_correlation_constant_not_forced.py::substrate_correlation_proximity_cap_is_contact_local_no_rank	56d7c16cfe

Appendix B: Dated history

Commits are found by exact subject in the remote history; times are UTC; a dagger marks a commit not contained in origin/dev.

TABLE III: The history of a_{corr} and R_{tail} in the corpus.

UTC	commit	subject
2026-08-11 13:52Z	`7a4f007840`	feat(abi): forward helium correlation -- the r12 term closes the mean-field residual to +0.42%
2026-08-11 18:18Z	`79ab032f5a`	feat(book): forward-derive the missing-terms constant a = pair-correlation energy
2026-08-11 18:23Z	`74bebdae80`	feat(book): the COMPLETE forward generator ; zero fitted constants, 2p row <1%
2026-08-11 21:27Z	`a4dc3e19eb`	fix: correct the E_c=a_corr over-claim ; verified they are different quantities
2026-09-17 02:30Z	`b2a7b01ba6`	feat(gr_qm): a_corr is measured, not derivable -- and the corpus disagrees with itself by 2
2026-09-18 01:32Z	`1ae99b7475`	feat(gr-qm): eps_2(0) does not close by the Green's-function assembly -- the crossover correction is non-elementary
2026-09-18 05:31Z	`5fbbc005ff`	research(gr-qm): prove the rigid-limit source non-elementary; the substrate answer is the deformed metric
2026-09-18 05:59Z	`0fc13294cf`	research(gr-qm): the substrate a_corr is a global field energy; the saturation root resums globally
2026-09-18 07:08Z	`b99c392aa1`	research(gr-qm): name the two-center Born-Infeld field energy as the open a_corr residual
2026-09-18 14:50Z	`674a28b5fd`	research(sympy): land a_corr coincidence match derivations and two open verdicts
2026-09-18 15:58Z	`caf6916754`	research(sympy): land proximity-cap contact-local no-rank verdict for a_corr
2026-09-18 17:19Z	`4626c0041a`	research: land two-center Born-Infeld field energy R_tail closed-form bracket
2026-09-18 20:06Z	`a2ea24ca21`	research: prove the two-center Born-Infeld field energy R_tail is non-elementary
2026-09-19 01:27Z	`6744384080`	research: downgrade R_tail two-center BI structure to a rank-6 upper bound with open obligations
2026-09-19 03:24Z	`745293d9c5`	fix(physics): derive R_tail trial bounds and correct correlation claims
2026-09-20 12:49Z	`de463350ee`	fix(gr-qm): reconcile conditional atomic energy and ionization
2026-09-20 23:12Z	`82ead9b8d9`	fix(gr-qm): consume exact relaxed occupation certificates
2026-09-21 05:09Z	`409d83bbff`	feat(gr-qm): two-center Born-Infeld minimizer is unique and two-sided bracketed
2026-09-21 14:34Z	`35a424b90d`	docs(gr-qm): label the lambda-reduction as the frozen superposition energy, not the true R_tail
2026-09-21 15:10Z	`826a804f28`	feat(gr-qm): validate the M11 true-minimizer route to a_corr - OPEN, blocking antecedent named
2026-10-28 23:16Z	`33b095be04`†	docs: record acorr derivation history and remaining antecedents
2026-11-01 03:55Z	`ef75344b6c`	feat(gr-qm): establish selected physical energy order lower bound
2026-11-01 12:06Z	`e3b4470853`†	feat: derive order17 frozen energy annihilator
2026-11-01 13:13Z	`9188e5a275`†	feat: classify order17 apparent divisors
2026-11-01 14:03Z	`703d37e793`†	test: independently check order17 divisor and endpoint data

Appendix C: Premise ledger

Consumed by the definition and the imported numbers: the nonrelativistic Coulomb comparison problem and its $1/Z$ expansion (imported reference model); the published ϵ_2 and per-l increments (literature, imported). None of these is a premise of the required derivation from the Born-Infeld Lagrangian and the particle equations; they enter only as cross-check oracles. Consumed by the static two-centre results: the Born-Infeld energy density (the chain's axiom), unit point sources in static Euclidean space (definitions of the diagnostic) and the finite-energy admissible class (commitment). Registered commitments that bound the conclusions: the M11 identification of the static two-centre energy with a_{corr}, and, for the true large-separation law, the induced-dipole structure. The $E_8 \times E_8$ geometry,

its D_8 placement and the particle equations are retained in the framework and are not consumed by these calculations.

---

[1] E. Gable, The static two-centre Born-Infeld problem in the substrate framework (2026), companion paper, papers/two_center_born_infeld.

[2] M. Born and L. Infeld, Foundations of the new field theory, Proc. R. Soc. Lond. A **144**, 425 (1934).

[3] A. Dalgarno and J. T. Lewis, The exact calculation of long-range forces between atoms by perturbation theory, Proc. R. Soc. Lond. A **233**, 70 (1955).

[4] P.-F. Loos and P. M. W. Gill, Correlation energy of two electrons in the high-density limit, J. Chem. Phys. **131**, 241101 (2009).

[5] C. Wang, Rates of convergence of the partial-wave expansion beyond the Kato cusp condition, Phys. Rev. A **88**, 032511 (2013).

[6] C. Schwartz, Importance of angular correlations between atomic electrons, Phys. Rev. **126**, 1015 (1962).

[7] D. P. Carroll, H. J. Silverstone, and R. M. Metzger, Piecewise polynomial configuration interaction natural orbital study of $1s^2$ helium, J. Chem. Phys. **71**, 4142 (1979).

[8] T. Kato, On the eigenfunctions of many-particle systems in quantum mechanics, Commun. Pure Appl. Math. **10**, 151 (1957).